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TRILINOS CORE PRODUCTS UPDATE

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KOKKOS



- See full release notes at:
<https://github.com/kokkos/kokkos/blob/develop/CHANGELOG.md>
- Release 5.0.X
 - Requires C++20
 - Support for C++20 Modules
 - Defaults to new Kokkos::View implementation (std::mdspan)
 - Should be fully transparent
 - Backported internal implementations for C++17 and C++20
 - Removed tasking and Makefile support
 - Support for multiple cmake languages in one build: Enable via
Kokkos_ENABLE_MULTIPLE_CMAKE_LANGUAGES=ON

	Kokkos 4.x	Kokkos 5.0
GCC	8.2.0	10.4.0
Clang (CPU)	8.0.0	14.0.0
Clang (CUDA)	10.0.0	15.0.0
IntelLLVM (CPU)	2021.1.1	2022.0.0
IntelLLVM (SYCL)	2023.0.0	2024.2.1
NVCC	11.0	12.2.0
NVC++	22.3	22.3
ROCm	5.2.0	6.2.0
MSVC	19.29	19.30

Various type alias deprecations and replacements in View

- ▶ `HostMirror` → `host_mirror_type`
- ▶ `scalar_array_type` → `data_type`
- ▶ `const_scalar_array_type` → `const_data_type`
- ▶ `non_const_scalar_array_type` → `non_const_data_type`
- ▶ `array_type` → `type`
- ▶ `DynamicView::array_type` → `DynamicView::uniform_type`

KOKKOS CONTINUED



- Improved first-touch handling for View initialization on host-parallel backends
- Added support for type conversions between simd types
- **CUDA**: improved performance of 128-bit CAS device atomics (60x on H100)
- **OpenMPTarget**: deprecated (lower perf than native backends, not feature complete, almost no users, no institutional support, ...). **OpenMP** backend is still fully supported!
- **OpenACC**: now supports:
 - Kokkos_Random
 - partition_space API
 - Custom scalar types, custom reducers
- **HIP**:
 - Fixed performance regression in 4.6 with lightweight kernels and parallel_reduce
 - **Avoid ROCm 7.1 if possible** – you may get incorrect results!

SACADO AUTOMATIC DIFFERENTIATION PACKAGE



- Refactor of Kokkos::View to use std::mdspan had a significant impact on sacado
 - std::mdspan design was influenced by Sacado requirements for extended data types
- Resulted in a total rewrite of Sacado (and Stokhos) internals
- Eliminated 10K lines of code in Sacado (still there for backwards compatibility with old view)
- Breaking changes:
 - Anything related to CacheFad is not supported in new impl
 - Assignment of View<Fad<Scalar>> to View<Scalar> or vice versa does not work
 - Can't rely on the return type of common_view_alloc_prop
- Other potential upcoming changes:
 - **LayoutContiguous** support may change (become a tag/variant of the Fad type itself)
 - **Constructors** for View<Fad> will have new mechanics for setting the derivative array size (not going to be just adding an extra dimension)
 - Support for **SYCL**: flat parallelism on a branch, hierarchic being worked on

TEUCHOS, THYRA AND GALERI



Teuchos

- MPI communication routines are wrapped in Kokkos Profiling regions: <https://github.com/trilinos/Trilinos/pull/14363>
- The CommandLineProcessor can now be used to harvest information about the runtime environment by passing --print-system-info: <https://github.com/trilinos/Trilinos/pull/14494>
 - env flags, ldd, modules loaded, mpi info, [nvidia/rocm]-smi, ...
- Case sensitive option to string validators in ParameterList
- Magistrate serialization support for StackedTimer

Thyra

- Solver/preconditioner routines can now directly be used with Tpetra objects without having to wrap them to Thyra: <https://github.com/trilinos/Trilinos/pull/14459>

Galeri

- The matrix generation was rewritten to use Kokkos - runs on device and is more performant: <https://github.com/trilinos/Trilinos/pull/14120>

TRILINOS NOW SUPPORTS THE MAGISTRATE TPL



- Magistrate is a powerful serialization library
 - Use for MPI and restarts
 - Invasive and non-invasive approaches
- Enable with:
Trilinos_ENABLE_magistrate=ON
- Make sure to pause StackedTimer before serialization!
- Package support:
 - Teuchos::StackedTimers
 - Coming soon: Tpetra, Phalanx and Panzer
- StackedTimer now contains aggregated data across restarts!

```
Teuchos::StackedTimer timer("Total Time");  
...  
auto list_of_running_timers = timer.stopAllTimers();  
magistrate::serializeToFile(timer, "my_app.restart");  
timer.startTimers(list_of_running_timers);
```

```
auto timer =  
magistrate::deserializeFromFile<StackedTimer>("my_app.restart");
```

```
auto buffer = magistrate::serialize(timer);  
  
auto d_timer = magistrate::deserialize<StackedTimer>  
(buffer->getBuffer());
```

<https://github.com/DARMA-tasking/magistrate>

MAGISTRATE EXAMPLE

- Handles inheritance with extra macro
- Three serializers:
 - Sizing
 - Packing
 - Unpacking
- **Invasive**: serialize function on class
- **Non-invasive**: stand alone `serialize(s, my_obj)` function
- Supports many **std** and **Kokkos** containers
- NOTE: Teuchos has a separate serialization capability for Tpetra **Scalar** type communication. No plans to replace.

```
class BaseTimer {
public:
    ...
#ifdef HAVE_TEUCHOSCOMM_MAGISTRATE
    template<typename Serializer>
    void serialize(Serializer& s)
    {
        if (s.isPacking()) {
            TEUCHOS_ASSERT(!running_);
        }
        s | accumulation_
          | accumulationSquared_
          | count_started_
          | count_updates_;

        if (s.isUnpacking()) {
            running_ = false;
        }
    }
#endif
    ...
};
```



TPETRA



- Performance improvements for creation of column maps:
<https://github.com/trilinos/Trilinos/pull/14902>
<https://github.com/trilinos/Trilinos/pull/14908>
- Improved on-device sorting of graphs and matrices (2x speedup on fillComplete on H100):
<https://github.com/trilinos/Trilinos/pull/14890>
- New copyAndPermute code path for graph transfers (toggled by setting the environment variable TPETRA_USE_NEW_COPY_AND_PERMUTE=ON): 10x speedup on GPU!
- Use of unique MPI tags for each round of communication
- The new "lalltofewv" communication option obtains speedups for many-to-few communication:
<https://github.com/trilinos/Trilinos/pull/14146>
- Tpetra can now leverage communication routines from MPI Advance:
<https://github.com/trilinos/Trilinos/pull/14169>
- A reordering of steps during data transfer that should avoid unexpected messages can be used by setting the environment variable TPETRA_GRANULAR_TRANSFERS=ON:
<https://github.com/trilinos/Trilinos/pull/13947>
- The Tpetra assembly examples were updated to better showcase current Kokkos practices:
<https://github.com/trilinos/Trilinos/pull/14430>
- Helper routines for the common use case of generating matrices by applying a filter to the entries of another matrix: <https://github.com/trilinos/Trilinos/pull/14583>

H100	1 rank	4 ranks
develop	12.1s	4.44s
this PR	2.65s	0.78s

Too many changes to list, see: <https://github.com/sandialabs/seacas/releases>

Highlights

- Greg Sjaardema retired!
- New SEACAS lead: Tolu Okusanya
- Change default setting for `lowerCaseVariableNames` to `false`
- EXPLORE: allow names or ids in list and select commands
- IOSS: add an S3 database component
- More robust error checking and reporting for parallel io consistency checks

ZOLTAN AND ZOLTAN2



Zoltan

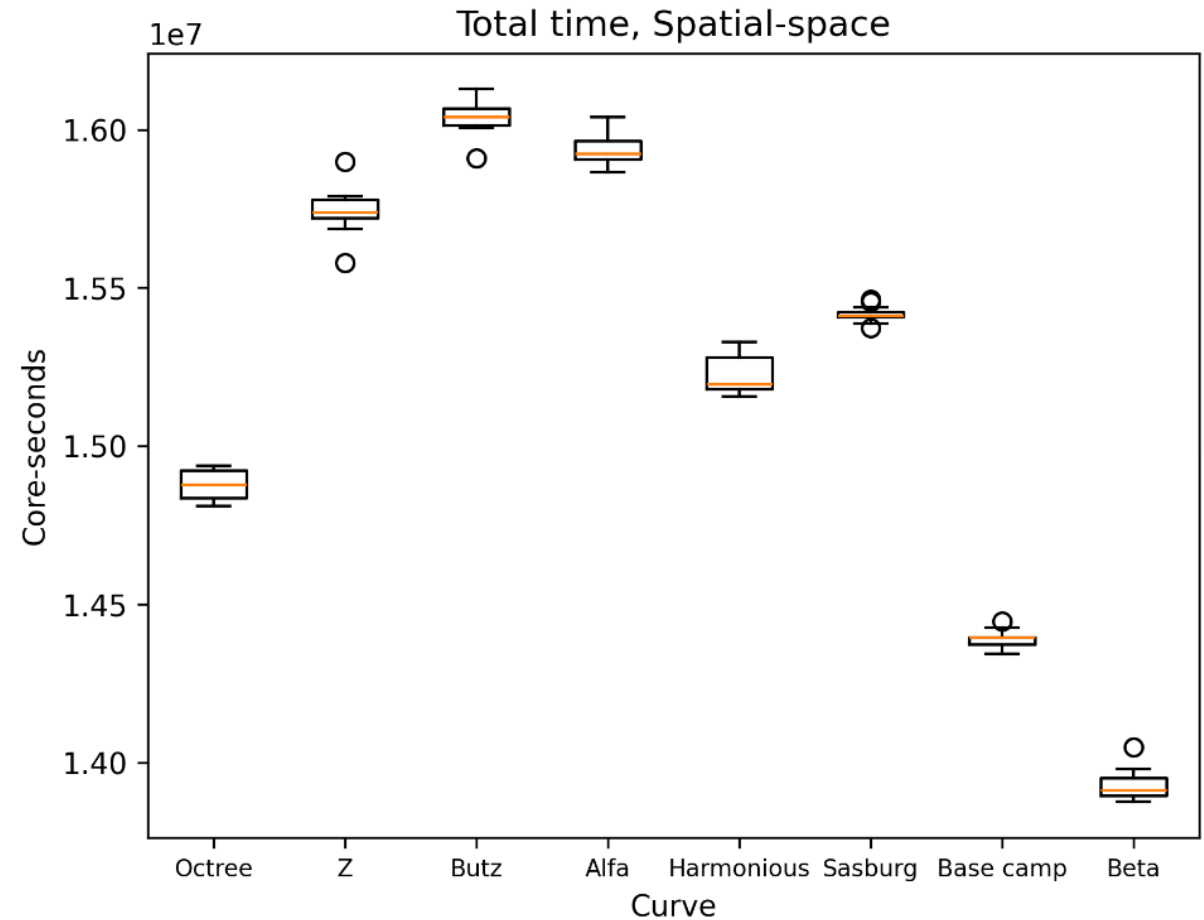
- Additional Hilbert Space filling curves:
<https://github.com/trilinos/Trilinos/pull/13880>

Zoltan 2

- Graph partitioning now supports complex-valued (Tpetra) CRS matrices
- Reverse Cuthill-McKee can now symmetrize the input matrix:
<https://github.com/trilinos/Trilinos/pull/14584>
- Epetra support removed

Future Plans

- Jet multi-GPU partitioner scheduled for integration into Zoltan2 (FY27)





END